

General linear model analysis

of fMRI data

Analysis phases

- Single-subject - first level - fixed effects analysis (FFX)
- Group - second-level - random effects analysis (RFX)
- (also mixed effects analysis is possible, but relatively rare due to large matrix size)

This is equivalent to e.g. conducting many trials per subject to measure reaction time, and then compute a subject-specific mean per condition, after which you would perform the actual statistical inference

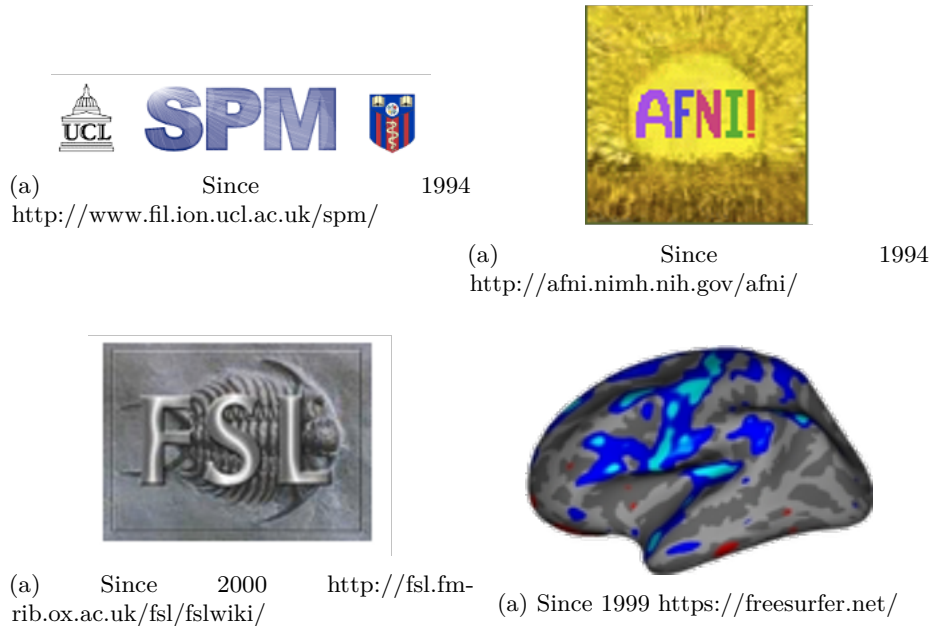
Multilevel modelling is also possible (for small datasets), but less frequent.

For a general overview of basic GLM analysis, see [Introduction to fMRI](#)

Before analysis

- Pick your analysis space, dictated by your hypotheses (see also [Preprocessing](#))
 - Individual volume (T1w)
 - Template volume (MNI*)
 - Individual surface (fsnative)
 - Template surface (fsaverage)
- Decide whether/how/how much to smooth the data

Software



Preprocessing used to be integrated into the software used for the actual analysis. And there are many software for this. On this slide I just put a few of them. Each of them has a slightly different philosophy behind it, but all of them do roughly the same thing, so at the end it should not matter much which of them you use.

One of the oldest ones are:

- SPM – statistical parametric mapping from UCL. It is a free MATLAB-based toolbox. However, it depends on the MATLAB licence, and has not been updated for a long time. The new update just came out which spans the gap from 2012 to 2025 (Tierney et al. 2025).
- AFNI - from NIH
- FSL from Oxford – a set of linux tools, also free and independent of MATLAB, which is an advantage, but requires Linux.
- FreeSurfer/FSFAST from the Martinos center - it is Linux based, and was originally conceived as a tool for structural data analysis, but has an fMRI module

Historical software

- broccolli, that can utilize GPU computing and can process the same dataset

n times faster

With the awareness of open science, and with the trend towards reproducibility and transparency, and with the release of fMRIPrep software, the preprocessing got detached from the actual data analysis.

Nilearn

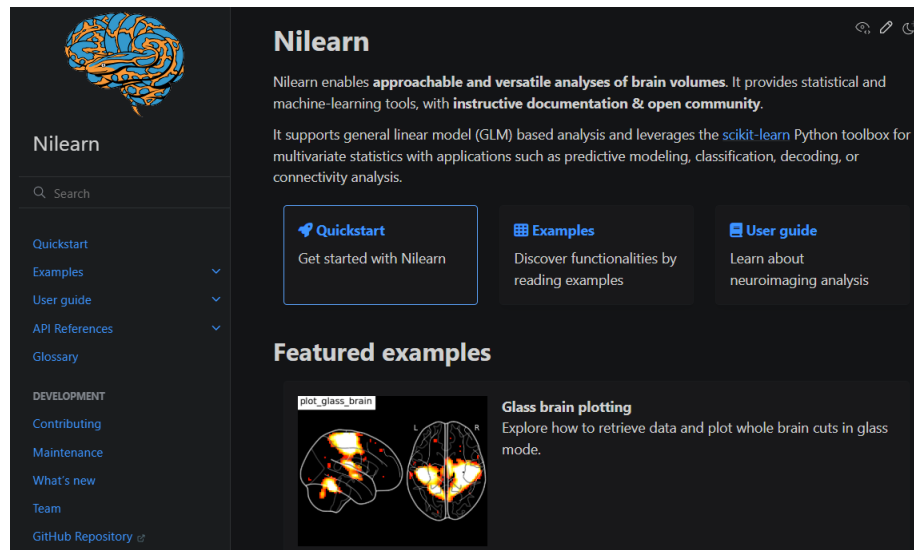


Figure 5: <https://nilearn.github.io/stable/index.html>

Ingredients

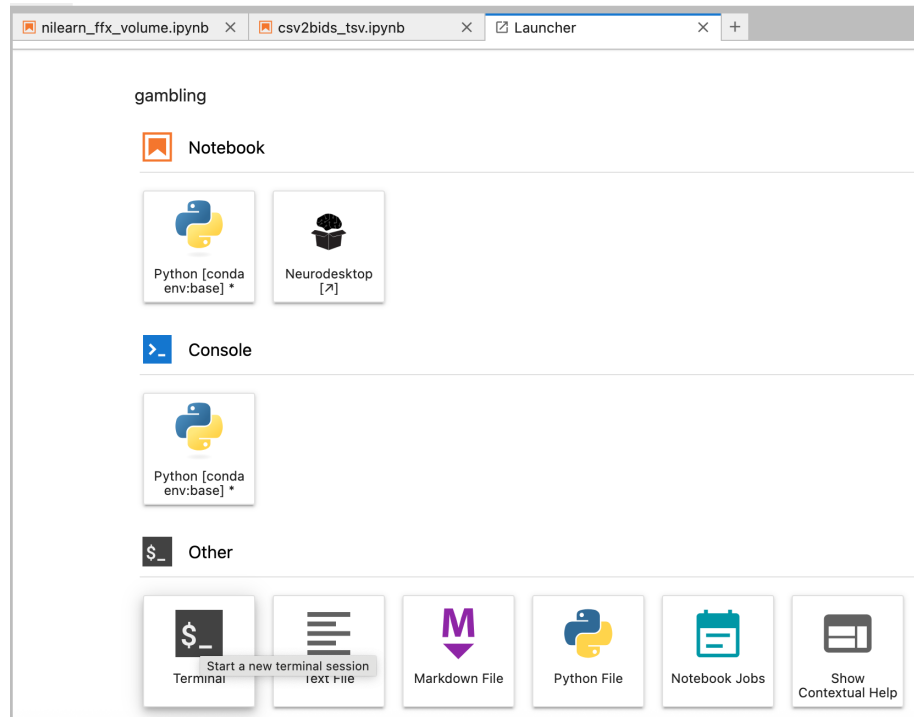
- fMRIPrep output
- events files in bids format, located in the same directory as the raw functional data
 - compare the psychopy csv files in “data” folder and the tsv bids files in the “bids” folder
 - you can copy them to your folder with the following terminal command or via the file manager

```
# gambling example  
rsync -a -v /shared/2025_SS_SE_ANI/gambling/tsv_behr_data /home/jovyan/gambling/
```

- A zip archive is also attached to the moodle assignment

Install nilearn

Open the terminal window



type:

```
pip install nilearn
```

and wait until the installation finishes

Run the analysis

- Open the jupyter notebook `nilearn_ffx_volume.ipynb`
 - you can copy it with the following terminal command or via the file manager

```
# gambling example  
rsync -a -v /shared/2025_SS_SE_ANI/gambling/nilearn_ffx_volume.ipynb /home/jovyan/g
```

- a copy is also attached to the moodle assignment
- try to run the analysis in the MNI space

Feel free to play around with the parameters and visualization options

References

Tierney, Tim M., Nicholas A. Alexander, Nicole Labra Avila, Yael Balbastre, Gareth Barnes, Yulia Bezsudnova, Mikael Brudfors, et al. 2025. “SPM 25: Open Source Neuroimaging Analysis Software.” <https://doi.org/10.48550/ARXIV.2501.12081>.